



Interpreting Deep-Learned Error-Correcting Codes: Experiments and Theoretical Aspects

Abhijeet Mulgund

University of Illinois Chicago

December 2, 2025



Natasha Devroye



György Turán



Miloš Žefran



Abhijeet Mulgund



Raj Shekhar

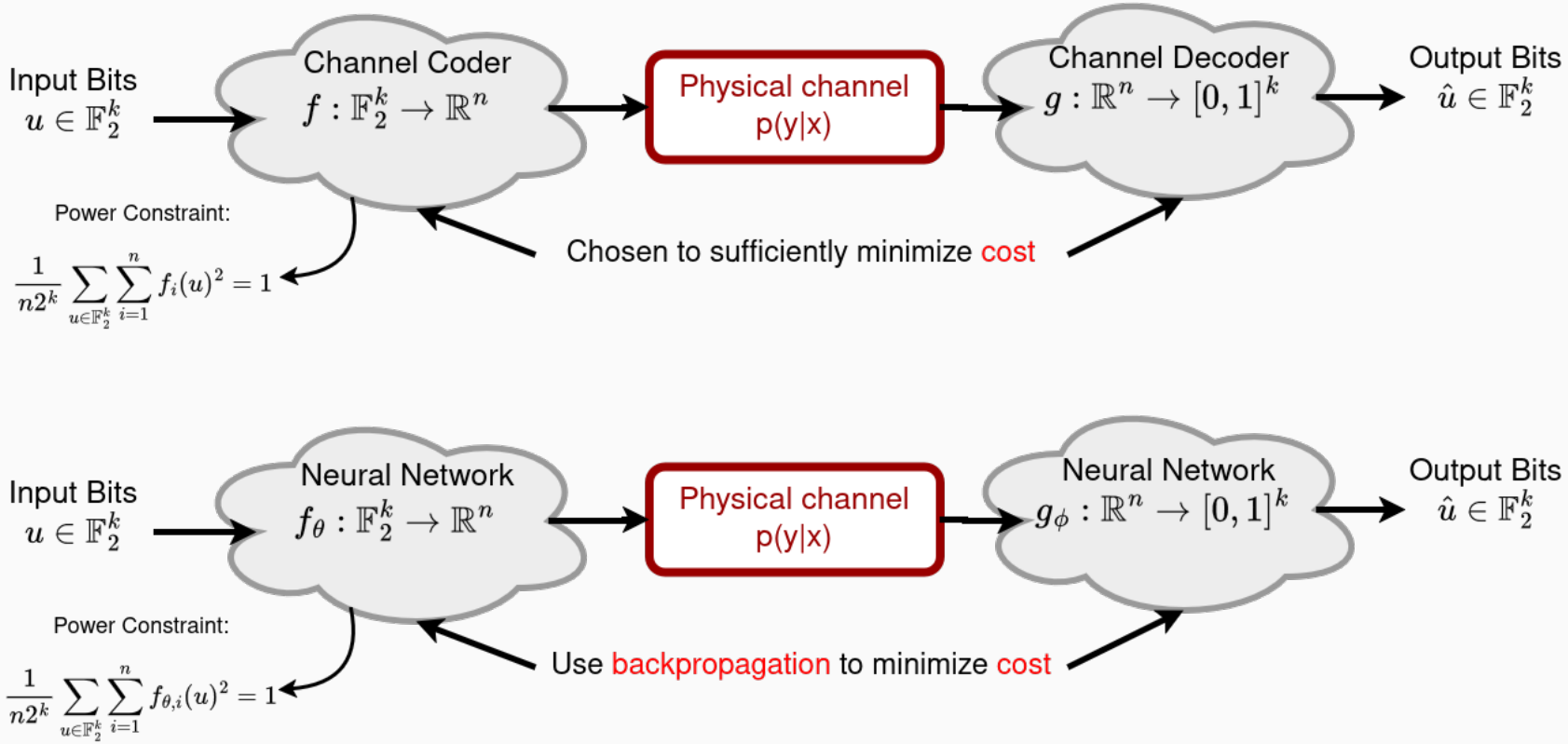


Yingyao Zhou

What are Deep-Learned Error Correcting Codes (DL-ECCs)?

General ECCs → DL-ECCs

Idea: Introduce redundancy and patterns into transmitted data so we can figure out how to undo corruptions during communication.



Examples: TurboAE [JKA+19], KO Codes [MLJ+21], Error-Correction Code Transformer [CW22]

Explaining DL-ECCs

Why Explain DL-ECCs?

Traditional Codes: Human designed, guided by theory and engineering insight - Interpretable!
DL-ECCs: Black-boxes, learned from data, somewhat inspired by traditional codes - Not Interpretable!

How can we explain what these black boxes learned?

Explanations provide several benefits:

- *Scientific Insight*: What can we learn from the DL-ECC?
- *Debugging*: Which aspects are "features" and which are "bugs"?
- *Robustness*: Performance guarantees under perturbed conditions?
- *Compression*: Extract a simpler model that uses same "algorithm"?
- *Sandboxing*: Mathematically rigorous problem; can we use DL-ECCs as a sandbox for XAI?

One-Slide Primer on Boolean Analysis

For $S \subset [k]$, the *parity* on S is the function

$$\chi^S(u) = \prod_{j \in S} (-1)^{u_j} = (-1)^{\bigoplus_{j \in S} u_j}.$$

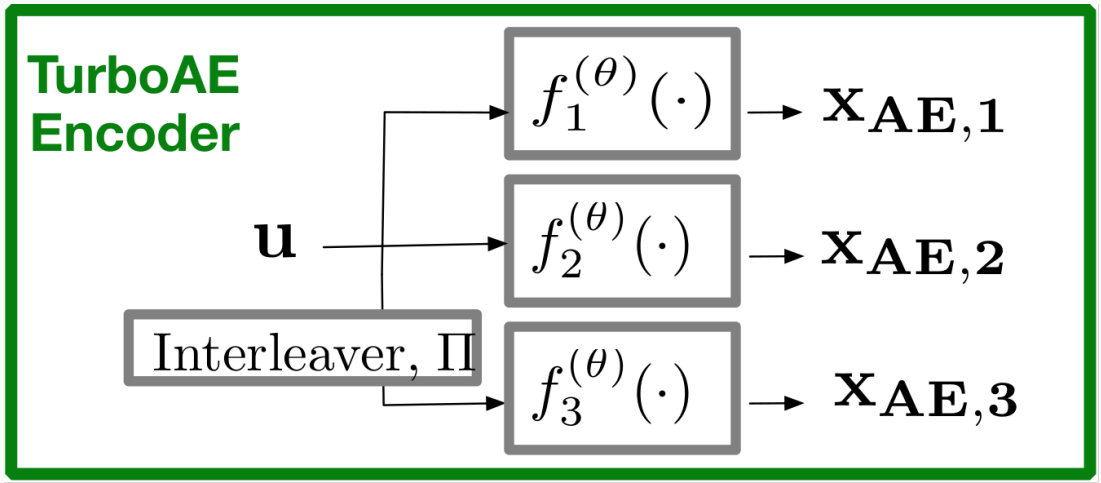
Any function $h : \mathbb{F}_2^k \to \mathbb{R}$ can be expressed as a *linear combination* of parity functions:

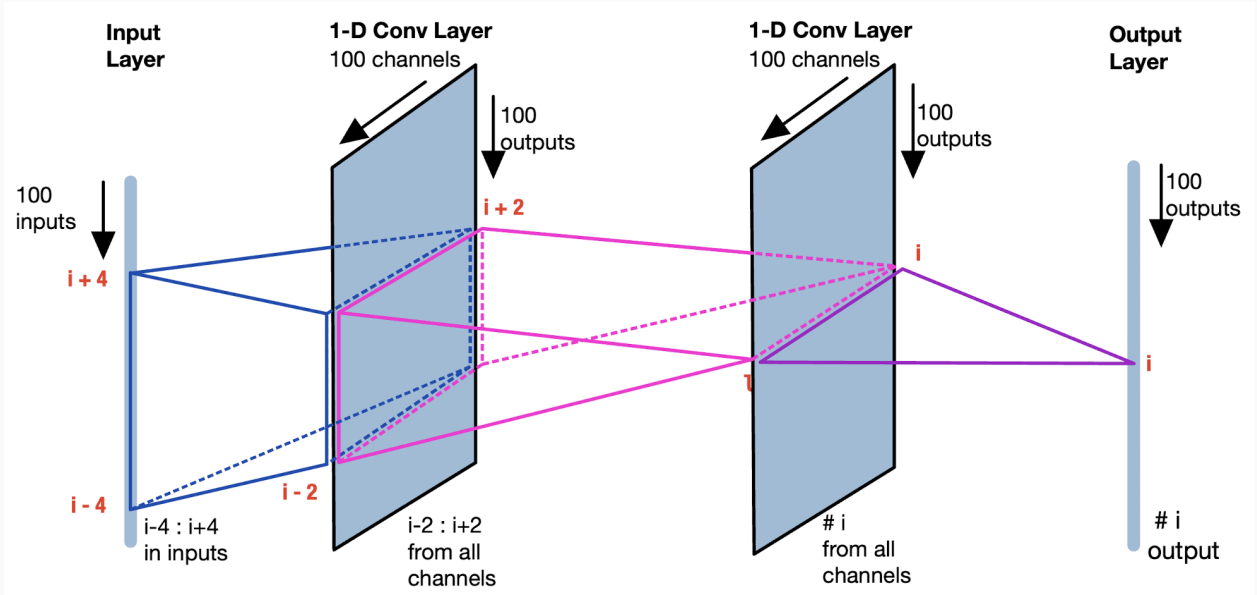
$$h = \sum_{S \subset [k]} \widehat{h}(S) \chi^S$$

called its Fourier decomposition. $|\widehat{h}(S)|$ measure how "close" h is to $\{\chi^S, -\chi^S\}$.

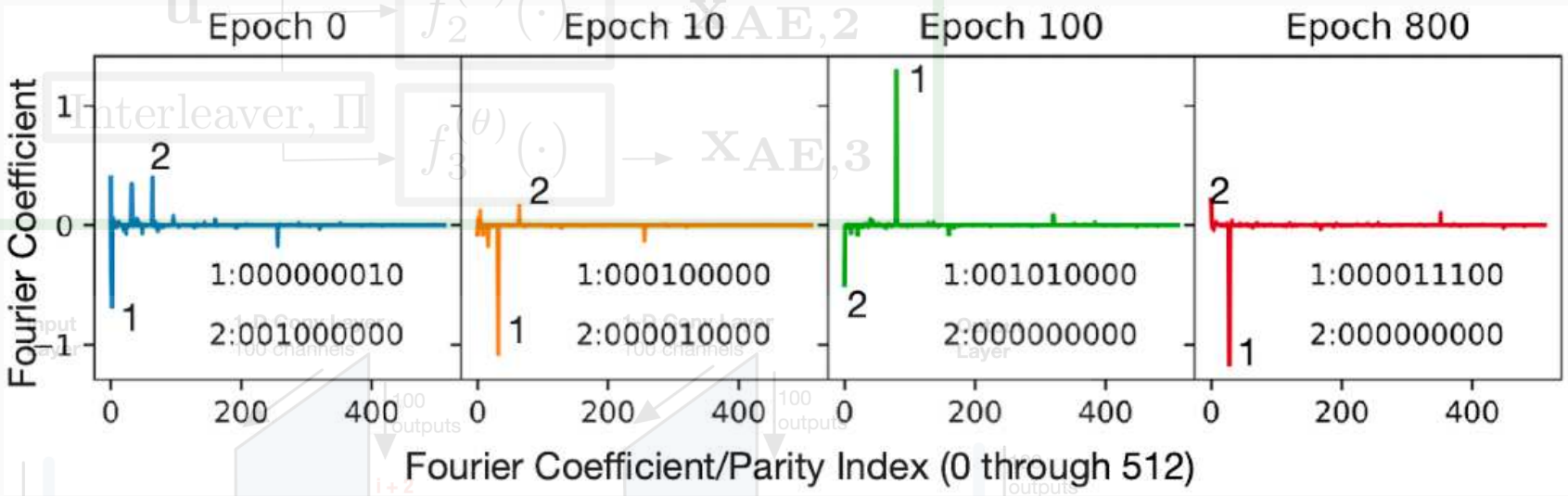
A *linear code* is a channel encoder $f : \mathbb{F}_2^k \to \mathbb{R}^n$ whose components are parities.

Empirical Observations: TurboAE Encoder



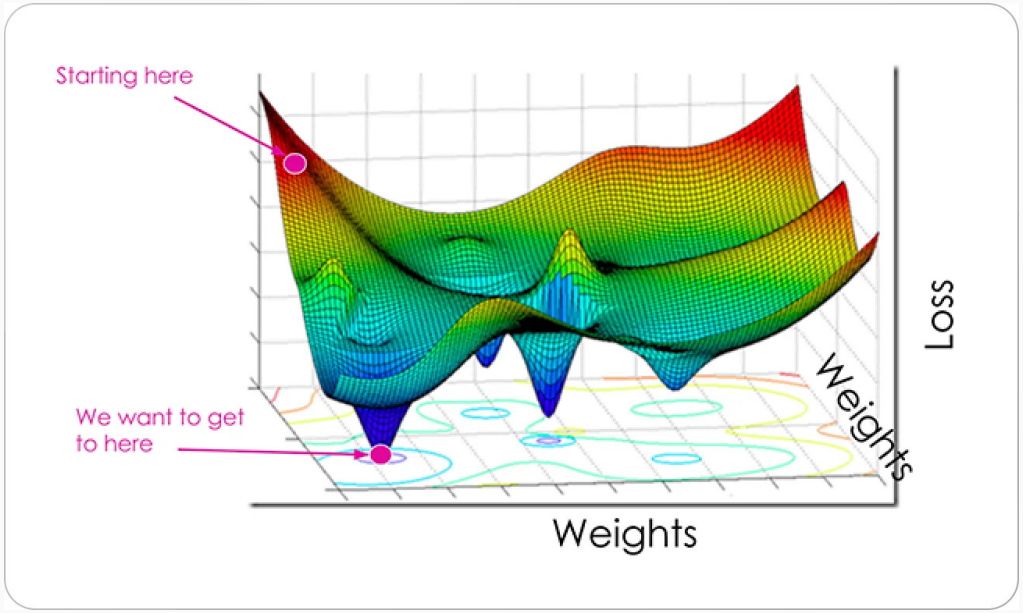


Empirical Observations: TurboAE Encoder



We also empirically saw in [MDTv23] solutions will be stuck in "saddles" corresponding to a dominant coefficient for most of training.

Loss Landscape

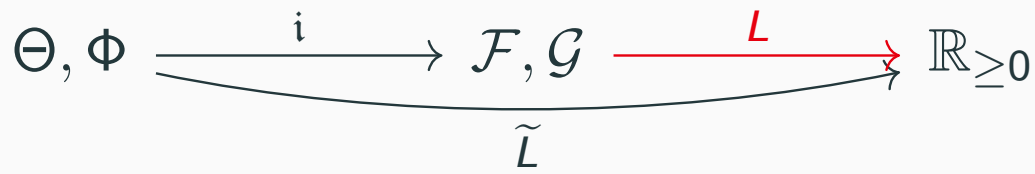


Source: Rosie Campbell

- How does the landscape govern what solutions DL-ECCs converge to?
- Can it explain some saddle-to-saddle behavior observed during training?
  - Is this behavior a consequence of using deep learning?

Spotlight: Loss Landscape of DL-ECCs

Key Idea: Decouple *problem induced* landscape from *neural network induced* landscape:



- Θ, Φ are the encoder, decoder network parameter spaces resp.
- $\mathcal{F}, \mathcal{G}$ are function spaces of encoders and decoders resp.
- $\mathfrak{i}$ is the *implementation* map: $\theta, \phi \mapsto f_\theta, g_\phi$
- L is loss on encoder-decoder pairs ($\widetilde{L} = L \circ \mathfrak{i}$ induced parameter loss)

Lemma (Informal)

Local minima (local maxima, critical points resp.) of L that are implemented by $\mathfrak{i}$ *pull back* to local minima (local maxima, critical points resp.) of $\widetilde{L}$ on parameters.

$\widetilde{L}$ inherits loss-landscape from L!

Spotlight: Loss Landscape of DL-ECCs

- channel $p(y|x)$ is a reflection-symmetric additive channel: $y = x + z$ for $z \sim q$ and q is invariant under coordinate reflections
- L is a "typical" bitwise loss metric (e.g. BCE, BER). In particular, for an encoder f, the optimal decoder is the MAP decoder $g_i(y) := p(u_i = 1|y)$, $i \in [k]$.

Theorem (Informal)

Let $n > 1$. For every non-constant linear code χ made up of parities $\chi_1, \ldots, \chi_n$, there exists "power allocation" $c \in \mathbb{R}^n$ so that $c \odot \chi := (c_1\chi_1, \ldots, c_n\chi_n)$ (with MAP) is a critical point of L and

$$L(c \odot \chi) \leq L(\chi).$$

Linear codes can attract gradient based algorithms!

Proof Idea

Application of ODE symmetry methods:

1. Identify invariances of L.
2. Note that invariances of L induce symmetries of the gradient flow ODE.
3. ODE symmetry methods apply uniqueness of solutions to construct symmetric solutions from initial conditions fixed by symmetries: *If the starting point of the gradient flow is a fixed point of a symmetry, the trajectory is always a fixed point of the symmetry.*
4. Finally apply compactness of the function space to get critical point existence.

Conclusion

Many takeaways and suggestions from this theorem

- **Scientific Insight**: Attempting to explain how the neural network arrived at its solution using loss-landscape analysis leads to novel *non-asymptotic* techniques for understanding error-correcting codes.
- **Debugging**: By theorem, there are $\Omega(2^k)$ distinct critical points, many of these will likely not be good. We can modify training procedure to more quickly escape these bad solutions. Heuristic methods for DL-ECCs seem to already do this!
- **Compression**: If we know our neural network is learning a "nearly" linear code, we can use an algorithm (e.g. Goldreich-Levin) to estimate which code that is.

Feel free to chat with me or email me at mulgund2@uic.edu for any thoughts, suggestions, and feedback.

Prior interpretability work for P2P DL-ECCs

- Performed for TurboAE [Jiang, Kim, Asnani, Kannan, Oh, Viswanath, NeurIPS 2019] in this sequence of papers:
  - N. Devroye, N. Mohammadi, A. Mulgund, H. Naik, R. Shekhar, Gy. Turán, Y. Wei and M. Žefran "Interpreting Deep-Learned Error-Correcting Codes," ISIT 2022
  - A. Mulgund, R. Shekhar, N. Devroye, Gy. Turán, and M. Žefran, "Evaluating interpretations of deep-learned error-correcting codes," Allerton 2022
  - N. Devroye, A. Mulgund, R. Shekhar, Gy. Turán, M. Žefran, Y. Zhou, "Interpreting Training Aspects of Deep-Learned Error-Correcting Codes," ISIT 2023
  - A. Mulgund; N. Devroye; Gy. Turán; M. Žefran, "Decomposing the Training of Deep Learned Turbo codes via a Feasible MAP Decoder," ISTC 2023
- Performed for KO Codes [Makkuva, Liu, Jamali, Mahdavifar, Oh, Viswanath, ICML 2021]
  - R. Shekhar; N. Devroye; Gy. Turán; M. Žefran, "Interpreting KO Codes," ISIT 2025

References


Yoni Choukroun and Lior Wolf, *Error correction code transformer*, Advances in Neural Information Processing Systems (S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, eds.), vol. 35, Curran Associates, Inc., 2022, pp. 38695–38705.


Yihan Jiang, Hyeji Kim, Himanshu Asnani, Sreeram Kannan, Sewoong Oh, and Pramod Viswanath, *Turbo autoencoder: Deep learning based channel codes for point-to-point communication channels*, NIPS, December 2019, pp. 2758–2768.


A. Mulgund, N. Devroye, Gy. Turán, and M. Žefran, *Decomposing the training of deep learned turbo codes via a feasible map decoder*, 2023 12th International Symposium on Topics in Coding (ISTC), 2023, pp. 1–5.


Ashok V Makkuva, Xiyang Liu, Mohammad Vahid Jamali, Hessam Mahdavifar, Sewoong Oh, and Pramod Viswanath, *KO codes: inventing nonlinear encoding and decoding for reliable wireless communication via deep-learning*, International Conference on Machine Learning, PMLR, 2021, pp. 7368–7378.